Joint Transmit Power and Rate Scheduling in Wireless Mesh Network for Performance Optimization ensuring Fairness

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Joint Transmit Power and Rate Scheduling in Wireless Mesh Network for Performance Optimization ensuring Fairness

Thesis submitted in partial fulfillment of the requirements for the degree of

Master of Technology

by

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To

My parents

DECLARATION

I declare that

a. The work contained in this thesis is original and has been done by myself and the general supervision of my supervisor.

b. The work has not been submitted to any other Institute for any degree or diploma.

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CERTIFICATE

This is to certify that the work contained in this thesis entitled "Joint Transmit Power and Rate Scheduling in Wireless Mesh Network for Performance Optimization ensuring Fairness" is a bonafide work of Subhrendu Chattopadhyay(Roll No. 124101008), carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.

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Abstract

Network performance optimization is a fundamental goal in Wireless Mesh Network (WMN). Effective scheduling and re-usability of space can significantly increase the performance of the system. Most of the modern wireless routers are equipped with Software Defined Radio (SDR) where radio properties like data rate and power settings are controllable. Proper adjustment of these two parameters can significantly increase thye spatial reuse capacity of the mesh network. However, in a Time Division Multiple Access (TDMA) network, proper scheduling strategy needs to be defined for successful exploitation of time and space reusability. Although improvement in network throughput is desirable, from end end user perspective, flow fairness needs to be ensured. This work studies the problem of finding an optimal and fair scheduling strategy, transmit power level and data rate selection a WMN bsead on IEEE 802.11s technology. However for successful communication SINR for each receiving stations need to ensured. This work designs a vector optimization problem for joint power and rate scheduling in IEEE 802.11s mesh network. The optimization problem maximizes network utilization and minimizes total transmit power required for communication. At the same time, fairness and minimum interference communication is ensured. The work also proves that, there exists a Pareto optimal solution for the proposed problem. However, the problem is known to be $\mathcal{NP}$-hard. So a distributed heuristic algorithm is proposed. Based on the distributed heuristic, the standard IEEE 802.11s channel access protocol is augmented to support the joint power and rate selection problem. Simulation results shows that the fairness of the system improves significantly. The proposed mechanism works well with respect to existing algorithm with out significant loss of throughput.

Contents

List of Figures

Chapter 1

Introduction

1.1 Background

From its inception Wireless Mesh Network (WMN) [1, 2, 3, 4, 5, 6] has been considered well suited for metropolitan area network (MAN) due to its cost efficiency, robustness and compatibility with traditional Ethernet or adhoc networks. In the last decade, a huge volume of research articles has been dedicated to this field of research. The main application of WMN is to provide broadband connection using multi-hop, multi-path wireless communication. IEEE 802.11 task group for wireless LAN provides the standard for WMN known as IEEE 802.11s [7]. As per the standard each WMN consists of two components called mesh station (mesh STA) and client station (client STA). Mesh STAs are the wireless routing entities which forms a backbone infrastructure for the client STA. Among Mesh STAs some of the mesh STAs act as gateways (mesh gate) which connects to other network segment or the Internet. Clients can connect to this backbone structure for accessing the Internet. Unlike STA, clients are devices with mobility. Mesh STAs are considered to have minimal mobility or none at all. Mesh STAs are collectively form Mesh Basic Service Set (MBSS) which uses the same mesh profile. Mesh functionalities are provided to only STAs that belongs to a MBSS. STAs under same MBSS forms wireless links between mesh neighbors. Even if two STAs are not in direct communication range, data can be forwarded by the other mesh STA. Fig. 1.1 represents an example of a WMN.

The standard for IEEE 802.11s describes protocol elements for mainly three different types of services named peer management, path selection and channel access. Peer management protocols help to maintain the mesh neighbors for a particular mesh STA. Path selection protocols provides layer-2 routing for WMN. Hybrid Wireless Mesh Protocol (HWMP) is one of the standard path selection protocol. IEEE 802.11s supported mesh STAs use different medium access control (MAC) for channel access. The primary focus of this work is on channel access techniques for improvement in network performance. A set of different channel access protocols are supported by IEEE 802.11 MAC. Following is a brief description about them.

1. Distributed Coordination Function (DCF): DCF is a contention based channel access mechanism. The access mechanism for DCF can be basic or reservation based. Basic access mechanism uses “Carrier Sense Multiple Access with Collision Avoidance” (CSMA/CA) with binary exponential back-off for collision resolution. For optional reservation based channel access DCF uses Request To Send/ Clear To Send (RTS/CTS) frames for contention free channel access through an entity called Network Allocation Vector (NAV). The use of NAV informs the mesh STAs about the reservation status of the mesh neighbors, thus providing Virtual Carrier Sensing (VCS). DCF is best suited for the delay insensitive

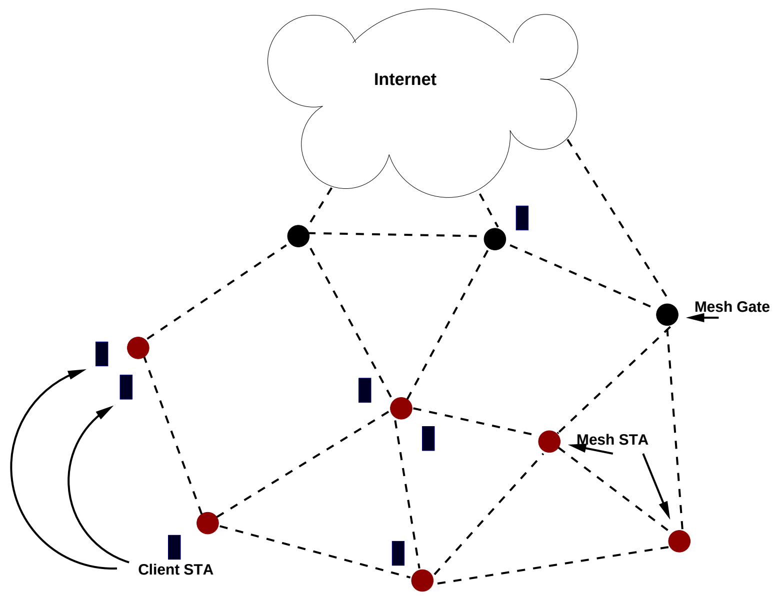

Fig. 1.1 Wireless Mesh Architecture

traffic. Due to its contention DCF sometimes performs poorly.

2. Point Coordination Function (PCF): PCF is a polling based contention free channel access mechanism. It is an optional protocol mentioned in the standard. PCF uses priority based VCS which is carried by periodic Beacon frames. PCF works better in the case of time bounded connection oriented services like voice and video traffic. Generally PCF and DCF works in a time sharing basis. Depending on the traffic class, for a fraction of time between two successive "Beacon" frame is dedicated for PCF. Rest are used as per DCF protocol. During PCF voice and video data packets are sent so that they get sufficient bandwidth. PCF requires all the mesh neighbors to perform PCF synchronously for smooth performance.

3. Mesh Coordination Function (MCF): MCF uses two different access protocol to ensure MAC layer Quality of Service (QoS) during channel access. This protocol uses mandatory contention based and optional contention free channel access mechanism by the following two protocol elements.

    (a) Enhanced Distributed Channel Access (EDCA): EDCA is a contention based access protocol. Based on different traffic classes acording to the Qos provisioning EDCA uses a variable contention window size. A mesh STA using EDCA can transmit multiple frames, if the the total transmission opportunity (TXOP) duration does not exceed TXOP limit.

    (b) MCF Controlled Channel Access (MCCA): MCCA is an optional feature in MCF to avoid contention during channel access. For avoiding contention during channel access, MCCA uses Spatial Time Division Multiple Access (STDMA) mechanism. However, not all mesh STAs in a mesh BSS are required to participate in MCCA. MCCA enabled mesh STAs tries to acquire channel using MCCA. Non MCCA enabled devices uses EDCA for channel access. Mesh STA scheduled in a TXOP by MCCA may need to participate in EDCA also for gaining channel from MCCA disabled

STAs. One advantage of using MCCA is to provide sufficient service guarantee to delay sensitive traffic.

The present work proposes enhancement on the top of the standard MCCA. MCCA uses

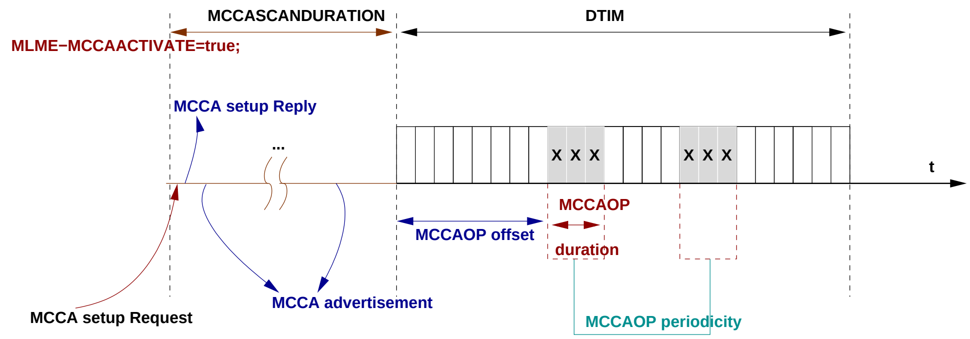

Fig. 1.2 MCCA Standard

periodic dynamic TDMA based channel access. Each period for MCCA is defined as Delivery Traffic Indication Message (DTIM) interval. DTIM interval is further slotted in MCCA opportunity (MCCAOP) slots. A reservation consists of three parameters as mentioned in Fig. 1.2. For the sake of simplicity let us consider that MCCAOP duration be σ and there exist a total of T number of MCCAOP per DTIM interval. Some STAs who wish to take part in data transfer first checks for any existing schedule. If the schedule is not available, it requests MCCA advertisement to mesh STAs, from which it has not received schedule. Let a mesh STA (MCCAOP owner) wants to send data packet to another mesh STA (MCCAOP responder). The control packet exchanges are shown in the Fig 1.3. In this context a STA If MCCA owner does not has any schedule information from any MCCA neighbor, it first sends MCCAOP advertisement req to the corresponding STA. After the mesh STA has schedules from all its mesh neighbors, it finds for a suitable schedule for its own data transfer. MCCAOP owner then sends a MCCA Setup Request as shown in the Fig. 1.2. This frame consists of 3 reservation parameters [MCCAOP offset, MCCAOP duration, MCCAOP periodicity]. Each MCCAOP responder maintains a map reservation table. Based on this map it sends a MCCA setup reply for either accepting or rejecting the schedule with an alternate reservation. If the schedule gets accepted then both MCCAOP owner and MCCAOP responder broadcast a MCCA advertisement frame which consists of all the tracked reservation in the sender STA and Mesh Access Fraction (MAF). MAF determines the ratio of reserved time by a STA over the DTIM time period. Upon receiving reject reply of MCCA setup, MCCAOP owner tries to re-calculate the MCCA reservation parameters. In this case recalculation may depend on the received alternate schedule. The throughput performance of a WMN often depends on several interdependent factor like physical layer data rate, transmit power as well as interference level. Some of them are investigated in the following section.

1.2 Motivation of the Thesis Work

In case of WMN one of the fundamental objective is to optimize the network utilization in terms of throughput. To achieve theoretical bound of overall network throughput, several works have been proposed in literature. The Work of Chakraborty et.al.[8] increases spatial reuse capability by enabling simultaneous transmission and reception. However, the effect of interference can be controlled by assigning different time slots for different interfering STAs. An practical approach

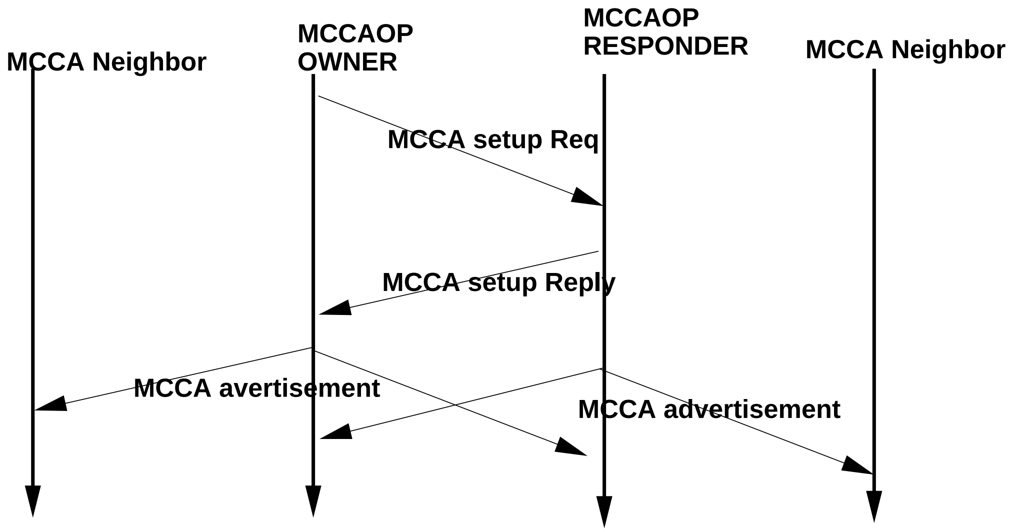

Fig. 1.3 MCCA Setup Procedure

for doing this is MCCA which ensures Spatial-TDMA. A STA equipped with Software Defined Radio (SDR) can tune its transmit power and data rate from an available set of configuration. Use of higher transmit power increase the chances of successful delivery of frames as Signal to Interference Noise Ratio (SINR) of the frame in the receiver STA increases. The increase in signal power limits the spatial reusabilty of the system as neighbor STAs are also interfered by the transmission. Thus decrease in overall network performance is observed. The reverse is true for increase in data rate. To increase data rate each STA uses some Modulation and Coding Scheme (MCS). It has been found that MCS providing higher data rate needs high receive sensitivity for successful reception/decoding of frames [9]. As a result to achieve higher data rate, transmit power needs to be higher also. But increase in transmit power also increases the interference level in the system. Thus packet error rate also increases. Therefore choosing a proper scheduling of transmit power level along with proper data rate to gain optimum throughput is a challenge in case of WMN. It has been studied in [10] that network performance in terms of throughput has a trade-off with fairness of the system. Hence it is challenging to find a strategy which provides fairness for Joint Power and Rate Scheduling (JPRS). In their paper [11] Li et.al have shown that optimal power control strategy with scheduling for only two discrete power levels is also $\mathcal{NP}$-hard because of the additive/cumulative interference effect on the receiver node. In [12] a distributed heuristic named Distributive Power control Rate adaptation Link scheduling (DPRL) has been proposed. DPRL effectively tries to solve power and rate scheduling iteratively. However, the work does not encompasses fairness criteria.

Though improvement in network throughput is desirable, providing fairness is also important for network end user perspective. For a fair system, the end users experience better Quality of Service (QoS). The term fair has different notions in the literature. If achieved throughput of each STA is proportional to the traffic load of the STA then it is called *proportionally fair*. Similarly in case of *max-min fair* the objective tries to maximize the throughput of minimally fair STA. It is debatable to use *max-min fairness*. Also being a global property *proportional fairness* is difficult to achieve in distributed environment. Jeonghoon *et.al.* [13] used a generalized fairness scheme named $(\mathfrak{P}, \alpha)$-Proportionally fair. According to them

Definition 1.1 *Let* $\mathfrak{P} = \{\mathfrak{P}_{11}, \mathfrak{P}_{12}...\mathfrak{P}_{nn}\}$ *and* α *be all positive numbers. Now a vector of rates*

$\mathscr{R}^*$ *is said to be* $(\mathfrak{P}, \alpha)$*-Proportionally fair if for any feasible vector* $\mathscr{R}$

$$\sum_{ij} \mathfrak{P}_{ij} \frac{\mathscr{R}_{ij} - \mathscr{R}^*_{ij}}{\mathscr{R}^{*\alpha}_{ij}} \leq 0$$

Their work also derived a function $F_\alpha(\mathscr{R})$ which has a property of achieving $(\mathfrak{P}, \alpha)$-Proportionally fairness. It is as following.

$$F_\alpha(\mathscr{R}) = \begin{cases} \mathfrak{P}_{ij} \log(\mathscr{R}) & \alpha = 1 \\ \mathfrak{P}_{ij} \frac{\mathscr{R}^{(1-\alpha)}}{(1-\alpha)} & Otherwise \end{cases} \tag{1.1}$$

The focus of this work is to provide fairness for joint power and rate scheduling and as well as optimize the performance. For ensuring fairness the proposed work considers $(\mathfrak{P}, \alpha)$-Proportional fairness because it generalizes the proportional fairness as well as max-min fairness. However, only power and rate control does not ensure fair allocation of the channel resources. For ensuring fairness of flows an explicit strategy needs to be employed so that each STA get fair share in throughput.

1.3 Contribution

In this work a centralized vector optimization problem is formulated for the optimization of total power consumption and packet delivery rate while ensuring fairness and minimun communication. Theoretical analysis over the centralized problem reveals that the problem is convex and Pareto optimal solution exists for the problem. However, the problem is known to be $\mathcal{NP}$-hard. Based on the property of the optimization a distributed heuristic algorithm is proposed. For practical implementation perspective standard MCCA mechanism is augmented. A simulation implementation for measuring performance of the algorithm is developed and it is compared with existing DPRL algorithm proposed in [12]. As mentioned earlier there is an inherent trade-off between fairness and throughput of a system. The simulation reveals that the gain in terms of fairness index is high. However, the proposed scheme performs approximately similar to DPRL in terms of throughput. The major contribution of this work can be summarized as following.

1. An multi-objective optimization problem is formulated for MCCA based channel access protocol. The solution of this problem results the assignment of transmit power and MCS along with scheduling parameters such that all links get $(\mathfrak{P}, \alpha)$-proportionally fair share of the throughput.

2. Mathematical analysis reveals that a Pareto optimal solution for the proposed optimization problem exists.

3. The problem is centralized in nature and finding solution is $\mathcal{NP}$-hard. Therefore a distributed heuristic is proposed exploring the properties of the centrlized problem.

4. Suitable adjustments are made for augmenting the standard MCCA for providing fairness based on the proposed algorithm.

5. Finally simulation are done to compare the performance with that of existing DPRL algorithm to show the improvement in fairness.

1.4 Organization of the Thesis

Rest of this thesis is organized as follows. Chapter 2 discusses recent studies done in power and rate control for WMN. Chapter 3 describes the network model and assumptions related to the scope of this work. Chapter 4 Chapter 5 mainly deals with the proposed solution approach and augmentation of standard MCCA protocol respectively. In Chapter 6 simulation results are given for comparison. Finally Chapter 7 concludes the thesis with future direction of work.

Chapter 2

Related Works

Initial works in this area of study used to consider only effective control of transmit power for throughput improvement. However there are some existing works on joint design strategies for different network parameters. A few existing works considered power and rate control based scheduling. However the author could not found any existing work that encompasses fairness with adaptive power and rate scheduling. In this chapter prior works are discussed.

Simple power control techniques proposed in the literature can be crudely classified into two subcategories.

2.1 Static Power Control

A Static Power Control strategy assigns transmit power level a priori. The nodes can not change their power levels on demand basis. Such scheme performs well where network topology does not change very often. Static power control mechanisms are often seen to give sub-optimal performance. However these protocols are not very suitable for networks with dynamic requirements. Based on the power level assignment, static power control techniques can be further sub-categorized into uniform and variable range policies.

2.1.1 Uniform Range Policy

A Uniform Range Policy uses single power level for all the nodes. Studies in [14, 15] have shown that throughput will be nearly optimal if all nodes are connected with minimum power level. Based on this observation, they have designed a power adaptation strategy, called COMPOW. Based on this technique, Gupta *et.al.*[16] suggested that the achieved best case capacity will be $\Theta(\frac{1}{\sqrt{n}})$ bits/sec where n signifies the nomber of nodes in the system. They have also shown that even in a random network without employing any complex technique near optimal achieved capacity could be up to $\Theta(\frac{1}{\sqrt{n \log n}})$. Though the COMPOW protocol is very simple, it has is own share of disadvantages also. In case of a highly clustered/grouped topology in spite of having small intra-cluster/ intra-group power level, inter-cluster/inter-group power level could be very high. Hence by the virtue of having equal power level the common usable power level becomes high also. Thus the nodes in the same cluster will face the effect of interference. This surely degrades the performance.

2.1.2 Variable Range Policy

Variable range policy assigns different power level to different nodes unlike Uniform Range Policy. To manage interference issue of COMPOW Kawadia *et.al.* [17] have adapted COMPOW for variable range power control. They proposed three different power control protocol - MINPOW,CLUSTERPOW and tunneled CLUSTERPOW. MINPOW protocol uses Bellman-Ford algorithm with power requirement as cost function. In case of CLUSTERPOW the the basic strategy used is that each node assigns itself a power level so that no other nodes in subsequent hops require to use a higher power level. The main difference in CLUSTERPOW and tunneled CLUSTERPOW is that the later one uses a recursive lookup mechanism for power assignment. Another approach based on minimal spanning tree (MST) exists in the literature, proposed in [18]. They also proved that the average traffic carrying capacity remains unchanged if more nodes enter the system. Variable range policy can out-perform uniform range policy. However, static power allocation does not perform well where multiple path can exist between any source and destination. Change of path forces to reschedule the transmit power of nodes, because the node scheduling changes the interference effect. Statically allocated power control can not cope up with this issue. In the case of mobile network this issue persists. Also minor changes in the network topology and parameter might lead to recalculation of power levels of all the nodes in the system, which can be costly for a large network.

2.2 Dynamic Power Control

In case of a network which undergoes a lot of change over the time, static power allocation is not beneficial. Hence the power allocation is often needed to be changed frequently. This decision of power change is made on per link, per slot or per packet basis. Power Assignment for Throughput Enhancement (PATE) [19] tries to avoid congested neighbor. A least congested neighbor is chosen for transmission such that it creates less interference with other already chosen nodes. In another algorithm, Power Control Multiple Access (PCMA) [20], use of separate control channel is assumed. This channel is responsible for sending “Busy Tone Signal” used for advertising of tolerance level. An similar study proposes Power Control Dual Channel (PCDC) [21]. In PCDC the control channel is used for interference tolerance level. Channel being a limited resource, use of separate control channel is not always advisable. Use of separate channel restricts the throughput gain on that particular channel. That is why, in their work, Muqattash *et.al.* proposed another new algorithm called POWMAC [22]. POWMAC uses a series of RTS/CTS control packet for negotiation of power assignment before actual data transmission. The limitation of this algorithm lies in ”power access window“ which can be under contention. To improve the performance, Akella *et.al.* [23] suggested enhancement where a sender decrements its power level without compromising with the data rate. In IEEE 802.11 MAC standard if a node is about to transmit it senses the carrier. The carrier sensing is done via measuring Received Signal Strength Indicator(RSSI). If the RSSI is found below a certain threshold called Carrier Sense Threshold, it starts the transmission. The value of carrier sense threshold and transmit power level together decides when and with what power will a node be involved in transmission. In their seminal work, Fuemmeler *et.al.* [24] proposed that this product of carrier sen threshold and transmit power level must be constant. The intuitive argument behind this, is a node with higher transmit power level will contribute more interference to others. Hence a node with high transmit power level must be more careful before sending. In [25], this idea is further exploited. According to their study, the spatial reuse can be increased by reducing transmit power or by increasing carrier sense threshold.

2.3 Joint Design Challenges

Power control along with rate adaptation mechanisms define topology of a network. As mentioned earlier a lot of problem like scheduling of nodes, ensuring fairness characteristics, routing etc. are interdependent with underlying topology of the multi-hop wireless network. Simple power and rate control algorithms can not capture these effects of different parameters. To address this interdependency, joint design approaches are proposed in the literature, as discussed next.

2.3.1 Joint Power Control and Scheduling

Joint design approach considers the interference relationship between links which are dependent on power level assignment of nodes. One the initial works in this direction was provided in[26], contains a two phase centralized protocol. Before each slot, each of the phase are executed. In the initial phase decision of maximum schedulable node set is made. This node set contains nodes which are certain distance apart to minimize effect of interference. In the subsequent phase, each node gets a power level so that SINR threshold does not get violated. In their work, Chen *et.al* [27] have proposed a two phase approach which is distributed in nature. In first phase all schedulable nodes measures the channel condition with a probe packet. Based on interference temperature metric some of them are scheduled. The links with low SINR value, are marked as undetermined and considered eligible for next scheduling decision. Second phase determines the power level for scheduled nodes. In this case an undetermined node can join the set for transmission, if they can, after assignment of optimal power levels. However their work does not consider the effect of rate adaptation and its dependency with the SINR threshold.

2.3.2 Joint Power Control and Routing

One interesting trade-off in multi-hop network is longer hop shorter path versus smaller hop longer path. Longer hop shorter path can deliver data with lesser amount of delay, however they are more probable to get interfered. On the other side, shorter hop ensures immunity to the interference, though the network delay will get affected. Works in [28] pointed out this trade-off effectively. In their work concept of loner link is also introduced. A link is called loner link if can not be scheduled with any other link in the network. Joint power control and routing algorithms tries to encompass this trade-off. The study of Cruz *et.al.*[29] presents a two phase protocol for routing, scheduling and power control.

2.3.3 Joint Power Control, Rate Control and Scheduling

Effective decoding of a received packet depends on mainly two parameters. If SINR of a received packet is greater than the receive sensitivity or SINR threshold of the transmitted MCS, then only a node can decode the packet successfully. Higher the data rate greater the SINR threshold. Notationally $r_1 < r_2 < ... < r_m$ then $\gamma(r_1) < \gamma(r_2) < ... < \gamma(r_m)$ where $\gamma(r_x)$ denotes SINR threshold for data rate r_x. For a transmission source i to destination j with rate r_x the relationship between SINR of a successfully received packet ($SINR_{pkt}$) is defined by Eq. (2.1)

$$SINR_{pkt} = \frac{P_{ij}G_{ij}}{\eta + \sum_{k \in I} G_{kj}P_{kj}} \geq \gamma(r_x) \tag{2.1}$$

Here P_{ij} and G_{ij} denotes the transmit power level and antenna and channel gain for a wirless link between i to j. η is the ambient noise. In their work Hedayati *et.al.*.[30] modeled the centralized integrated power and rate control problem in case of STDMA network using a Mixed Integer Linear Programming (MILP). They have shown that finding a Pareto optimal solution of the MILP problem is $\mathcal{NP}$-hard. They also reduced the problem to maximum independent set problem, and proposed a greedy heuristic solution to the problem. In the subsequent works [31, 12], they proposed distributed heuristic protocol named DPRL. DPRL protocol uses broadcasting of SINR value. The link with highest SINR margin wins and then it performs power and rate calculation based on the channel quality. This process iterates untill all the links are scheduled. However the proposed method suffers from some serious issues like like of fairness. Simulation study reveals that a link with high SINR margin is most likely to win in each round of STDMA super frame. Thus links with lower power margin suffers significantly due to unavailability of slots. Packet error rate for that link will be significantly higher due to cumulative interference effect. Hence for providing fairness, low power margined links needs more slots. Also the proposed method uses STDMA based scheduling which is not yet supported in the IEEE 802.11s standard.

Chapter 3

System Model

IEEE 802.11s MAC layer specification with IEEE 802.11b/g/n physical layer support uses multiple frequency channels for contention free channel access mechanism. However many community WMN uses single channel for mesh backbone because static channel assignment for a particular mesh-STA limits the neighbor nodes. On the other hand dynamic channel assignment is known to be $\mathcal{NP}$-hard and current heuristics available for channel assignment is costly. So significant changes in current standard is required. This work considers single channel single interface for the sake of simplicity. However the proposal can work with some modification with multi-channel and multi-interface conditions. In this context omni directional antenna is considered. The proposed solution works best in WMN backbone structure where mobility is negligible.

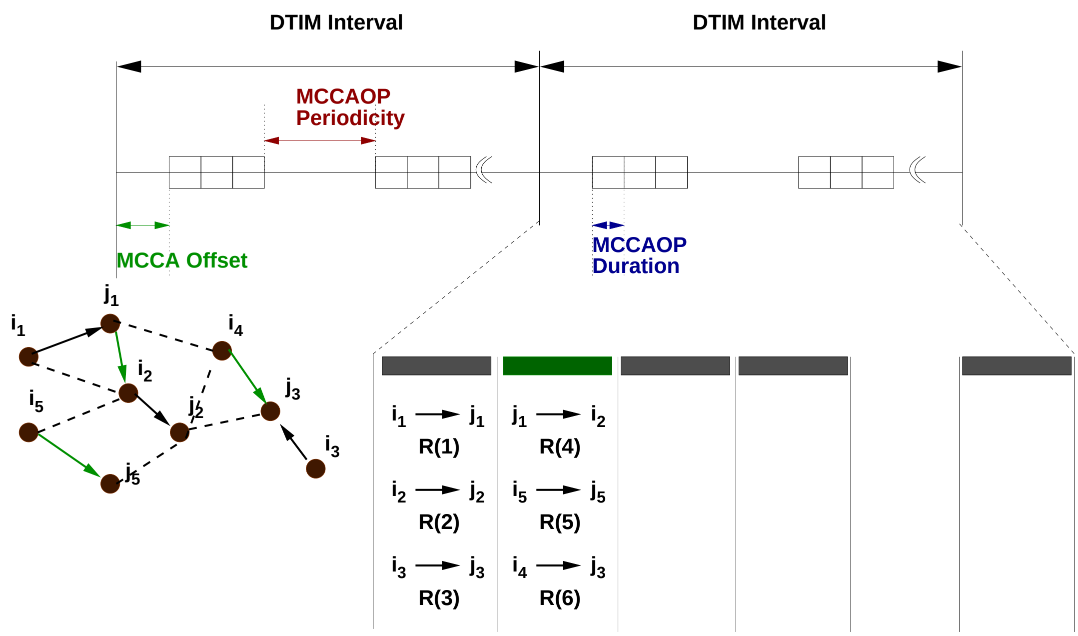

Fig. 3.1 MCCA Schedule Mechanism

In this work each flow is distinguished by it's source and destination id. A communication between source i and destination j is denoted as $i \rightarrow j$. However in a distributed environment it is hard to identify end to end flow. Hence for the simplicity each subflows are identified with a flow ID. Throughout the thesis r_h is considered as data rate for h-th MCS. Also consider each Mesh STA supports m different such MCSs. R denotes the set of all available data rates. So

$R = \{r_1, r_2 ... r_m\}$ such that $r_1 < r_2 < ... < r_m$. Sometimes the thesis adapts to the notation $i_k \underset{r_k}{\rightarrow} j_k$ to signify flow i_k to j_k with r_k data rate. The communication channel time is divided into identical channel slot (MCCAOP) of duration σ.

For simplicity the author further assumes presence of uniform power level support in MAPs. Let power level used by STA i for communication $i \rightarrow j$ at time t-th MCCAOP be denoted as $P_{ij}^{(t)}$. It is assumed that $\forall t \in \{1, 2...T\} : P_{ij}^{(t)} \in [0, P_{max}]$.

Let $X_{ijh}^{(t)}$ denotes the channel reservation matrix which has the following interpretation

$$X_{ijh}^{(t)} = \begin{cases} 1 & \text{If flow i} \rightarrow \text{j uses rate h at time t} \\ 0 & \text{Otherwise} \end{cases}$$

Notation Tx_{ij} is used for denoting rate of sent packets by flow $i \rightarrow j$ per DTIM interval

$$Tx_{ij} = pkt/DTIM = [MCCAOP_{ij} \times MCCAP_{ij} \times r_h \times \sigma] = \sum_{t}^{DTIM} \sum_{h} (X_{ijh}^{(t)} \times r_h \times \sigma)$$

where $MCCAP_{ij}$ denotes the periodicity of the corresponding flow.

For a sub-flow $i \rightarrow j$ antenna and channel gain is represented by G_{ij}. Through out literature it is commonly assumed that antenna and channel gain is homogeneous i.e. $G_{ij} \approx G_{ji}$ [21],[20],[26]. The reason behind this assumption is all wireless routers are equipped with similar type of interfaces. According to the principle a wireless node can successfully decode the signal withstanding a certain Bit Error Rate (BER) if the receiver of the signal receives the signal with atleast threshold SINR. This SINR depends on the selected MCS. Threshold for rate r_h is denoted by $\gamma(r_h)$.

A feasible transmission scenario $S(t) = \{i_1 \underset{R(1)}{\rightarrow} j_1, i_2 \underset{R(2)}{\rightarrow} j_2 ... i_k \underset{R(k)}{\rightarrow} j_k\}$ under power allocation vector $P(t) = \{P_{i_1 j_1}^{(t)}, P_{i_2 j_2}^{(t)} ... P_{i_k j_k}^{(t)}\}$ at time t means the SINR level in each receiver(j_x) must be over threshold($\gamma(r_x)$). This can better be explained mathematically by the following Eq. (3.1).

$$\frac{G_{i_k j_k} P_{i_k j_k}^{(t)}}{\eta + \sum_{x \neq k} G_{i_x j_k} P_{i_x j_x}^{(t)}} \geq \gamma(r_k) \tag{3.1}$$

So the power profile of each STA must ensure that the constraint is satisfied. However the notion of optimality is for power allocation is redefined in this context.

Definition 3.1 *A power vector $P(t)$ is defined to be Pareto optimal for a particular feasible transmission scenario $S(t)$ where any other feasible power assignment $P'(t)$ needs atleast as much power as $P(t)$ for each Mesh STA in the same transmission scenario $S(t)$. Mathematically*

$$\forall z \in \{1, 2...|P(t)|\} : P_{i_z j_z}^{(t)} \leq P_{i_z j_z}^{'(t)}$$

To achieve Pareto Optimal power allocation, this work proposes a Multi-Objective Integer Program (MOIP) which considers Power and Rate allocation along with scheduling strategy. However it have been observed that the link scheduling problem does not consider fairness issues. To achieve the fair share of time this work considers $(\mathfrak{P}, \alpha)$-Proportional fairness scheme as mentioned earlier. Using the fairness criteria the objective of the system modifies to achieve

Pareto Optimal Joint Power and Rate Scheduling which ensures $(\mathfrak{P}, \alpha)$-Proportional fairness. The problem tries to achieve the maximum fairness while using minimum transmit power level. Let this problem be called as Fair-JPRS problem. A sample solution scenario is depicted in Fig. 3.1. The nodes connected via dotted lines can interfere each other. For example $i_4 \rightarrow j_3$ and $i_2 \rightarrow j_2$ communication can not be scheduled simultaneously. A sample schedule is shown in the figure.

Chapter 4

Optimization Problem Formulation

In this section Vector Optimization Problem is proposed to solve the Fair-JPRS problem. The properties of the problem is explored to show the correctness proof. Also

4.1 Optimization Problem Formulation

The proposed Vector Optimization Problem (VOP) has two different aspects. First to schedule each station with a power vector that is optimal. Secondly each STA gets a fair share of the time slices. Let $\Gamma(\alpha)$ be an indicator variable such that

$$\Gamma(\alpha) = \begin{cases} 1 & \alpha = 1 \\ 0 & Otherwise \end{cases} \tag{4.1}$$

Then Eq. (1.1) can be represented as following

$$F_{\alpha}(Tx) = \mathfrak{P}_{ij} \left(\Gamma(\alpha) \log(Tx) + (1 - \Gamma(\alpha)) \frac{Tx^{(1-\alpha)}}{(1-\alpha)} \right) \tag{4.2}$$

For the sake of notational simplicity let us also consider

$$\mathscr{X}_{ij} = \{Tx_{ij}, P_{ij}\} \tag{4.3}$$

Clearly the objective functions for scheduling and power allocation is dependent upon $\mathscr{X}_{ij}$. Hence two objectives can represented in terms of $\mathscr{X}_{ij}$ as following.

$$Schedule(\mathscr{X}) = -\sum_{ij} (F_{\alpha}(Tx_{ij})) \tag{4.4}$$

$$Power(\mathscr{X}) = \sum_{ij} \sum_{t} \left(P_{ij}^{(t)} \right) \tag{4.5}$$

As mentioned earlier Eq. (4.4) is to ensure $(\mathfrak{P}, \alpha)$-Proportional fairness. Here $\mathfrak{P}_{ij}$ denotes the priority of the sub-flow $i \to j$ which can be determined by the service class of the flow. However required power vector needs to be minimized also simultaneously which can be represented as per Eq. (4.5). The negetive sign in Eq. (4.4) is used to keep homogeneity of the optimization objective(i.e Minimization). In course of optimization each flow must follow the SINR constraint.

$$\Phi[X_{ijh}^{(t)} - 1] - G_{ij}P_{ij}^{(t)} + \gamma(r_h)\sum_{fs} G_{fj}P_{fs}^{(t)} + \gamma(r_h)\eta \leq 0 \tag{4.6}$$

Here Φ is a substantially large constant which is used to eliminate redundant SINR constraints on sub-flows which are not scheduled at t. Also each mesh STA must be scheduled in such a way not more than one flow has same the receiver (Hidden Node Constraint). Following equation captures this constraint.

$$\sum_{h}\left[\sum_{ij} X_{ijh}^{(t)} + \sum_{jf} X_{jfh}^{(t)}\right] \leq 1 \tag{4.7}$$

Now the entire Vector Optimization can be expressed as following.

Problem 4.1 (Vector Optimization Problem)

$$Minimize \quad \mathscr{Q}(\mathscr{X}) = \{Schedule(\mathscr{X}), Power(\mathscr{X})\} \tag{4.8}$$

S.T:

$$0 \leq P_{ij}^{(t)} \leq P_{max} \qquad h \in \{1, 2...m\} \qquad t \in \{1, 2...DTIM\} \tag{4.9}$$

$$\sum_{h}\left[\sum_{ij} X_{ijh}^{(t)} + \sum_{jf} X_{jfh}^{(t)}\right] \leq 1 \tag{4.10}$$

$$\Phi[X_{ijh}^{(t)} - 1] - G_{ij}P_{ij}^{(t)} + \gamma(r_h)\sum_{fs} G_{fj}P_{fs}^{(t)} + \gamma(r_h)\eta \leq 0 \tag{4.11}$$

Next lets discuss about the equivalence of Fair-JPRS problem and formulated Vector Optimization Problem.

Lemma 4.1 *Every solution of the Problem 4.1 formulation yields a feasible transmission scenario at each time slot.*

Proof: Let $S(t), P(t)$ be a feasible solution of the Problem 4.1 for any arbitrary scenario and time slot t such that $S(t) = \{i_1 \underset{r_1}{\rightarrow} j_1, i_2 \underset{r_2}{\rightarrow} j_2, ...i_m \underset{r_m}{\rightarrow} j_m\}$ and $P(t) = \{P_{i_1j_1}^{(t)}, P_{i_2j_2}^{(t)}, ...P_{i_mj_m}^{(t)}\}$. The Eq. (4.10) guarantees that all transmitter and receiver are distinct in each time slot and thus in t also. This prevents hidden node scenario. Let $i_k \underset{r_k}{\rightarrow} j_k$ be any arbitrary transmission in $S(t)$ with power profile $P_{i_kj_k}^{(t)}$. Hence based on Eq. (4.11) following can be said

$$\frac{G_{i_kj_k}P_{i_kj_k}^{(t)}}{\eta + \sum\limits_{x \neq k} G_{i_xj_k}P_{i_xj_x}^{(t)}} \geq \gamma(r_k) \tag{4.12}$$

Eq. (4.12) suggests SINR constraints for successful transmission. This proves that $S(t), P(t)$ generates all feasible transmission scenario for all time slots. $\square$

With the statement of Lemma 4.1 the following can be said.

Theorem 4.1 *All optimum solutions of Problem 4.1 generates a Pareto optimal power vector allocation based on the transmissions scheduled in each time slot.*

Proof: Let $S^*(t)$,$P^*(t)$ be an optimum solution for Problem 4.1 at any arbitrary time slot t such that $S^*(t) = \{i_1 \underset{r_1}{\rightarrow} j_1, i_2 \underset{r_2}{\rightarrow} j_2, ...i_m \underset{r_m}{\rightarrow} j_m\}$ and $P^*(t) = \{P^{*(t)}_{i_1j_1}, P^{*(t)}_{i_2j_2}, ...P^{*(t)}_{i_mj_m}\}$. Say the Pareto optimal solution is $P'(t)$. To prove the Pareto optimality of $P^*(t)$ it is enough to prove that $P^*(t) = P'(t)$ component wise. Since there there exists only m simultaneous transmissions at t hence only m instances of Eq. (4.11) are non redundant. These active constraints can be expressed as following

$$G_{i_kj_k}P^{*(t)}_{i_kj_k} - \gamma(r_k)\sum_{x \neq k} G_{i_xj_k}P^{*(t)}_{i_xj_x} \geq \gamma(r_k)\eta \tag{4.13}$$

Now the set of constrains in Eq. (4.13) has a non-negative solution such that $P^*(t) \geq P'(t)$ component wise. This statement is can be proved by using Perron-Frobenius theorem. This result has been addressed in the works by Mitra [32], Hedayati et.al. [30] and Bambos et.al [33].

Again the optimality of $P^*(t)$ suggests the fact that $P^*(t) \leq P'(t)$. So one can conclude $P^*(t) = P'(t)$ $\square$

It can be shown that Problem 4.1 is $\mathcal{NP}$-hard. For that a simplistic form of the original problem is considered which incorporates the assumption of $\alpha = 0$, fixed power and fixed MCS. Also additive interference is ignored. If each sub-flow requires only one transmission opportunity then maximum independent set problem can be easily reduced to the scheduling problem for each time slot. As maximal independent set is a known $\mathcal{NP}$-Complete problem thus one can say so is Fair-JPRS problem.

4.2 Properties of VOP

In this section let us explore the convexity property of Problem 4.1.

Lemma 4.2 *Schedule($\mathscr{X}$) is differentiable under $\mathscr{X}_{uv}$ and a convex function.*

Proof:

$$\frac{\partial F_\alpha(x)}{\partial x} = \mathfrak{P}_x\left(\frac{\Gamma(\alpha)}{x} + \frac{(1-\Gamma(\alpha))}{x^\alpha}\right) \tag{4.14}$$

From Eq. (4.3)

$$\frac{\partial f(\mathscr{X})}{\partial Tx_{ij}} = \begin{bmatrix} \frac{\partial f(\mathscr{X})}{\partial Tx_{ij}} & \frac{\partial f(\mathscr{X})}{\partial P_{ij}} \end{bmatrix}$$

From Eq. (4.4), (4.2) and (4.14)

$$\frac{\partial Schedule(\mathscr{X})}{\partial \mathscr{X}_{uv}} = \begin{bmatrix} \frac{\partial Schedule(\mathscr{X})}{\partial Tx_{uv}} & \frac{\partial Schedule(\mathscr{X})}{\partial P_{uv}} \end{bmatrix} = \begin{bmatrix} -\mathfrak{p}_{uv}\left(\frac{\Gamma(\alpha)}{Tx_{uv}} + \frac{(1-\Gamma(\alpha))}{{Tx_{uv}}^{\alpha}}\right) & 0 \end{bmatrix} \tag{4.15}$$

Similarly

$$[H_s] = \frac{\partial^2 Schedule(\mathscr{X})}{\partial \mathscr{X}_{uv}^2} = \begin{bmatrix} \frac{\partial^2 Schedule(\mathscr{X})}{\partial Tx_{uv}^2} & \frac{\partial^2 Schedule(\mathscr{X})}{\partial P_{uv} \partial Tx_{uv}} \\ \frac{\partial^2 Schedule(\mathscr{X})}{\partial Tx_{uv} \partial P_{uv}} & \frac{\partial^2 Schedule(\mathscr{X})}{\partial P_{uv}^2} \end{bmatrix} = \begin{bmatrix} \mathfrak{p}_{uv} \left(\frac{\Gamma(\alpha)}{Tx_{uv}^2} + \frac{(1-\Gamma(\alpha))}{Tx_{uv}{}^{-(\alpha+1)}} \right) & 0 \\ 0 & 0 \end{bmatrix} \tag{4.16}$$

$Tx_{uv} > 0$ as it signifies data transmitted in each DTIM interval for each active $u \rightarrow v$ flow. $Tx_{uv} = 0$ Signifies $u \rightarrow v$ flow is inactive for a particular DTIM interval. $[H_s]$ positive semi-definite as it is symmetric and all its diagonal elements are non negative. □

Lemma 4.3 *$Power(\mathscr{X})$ is differentiable under $\mathscr{X}_{uv}$ and is a convex function.*

Proof:

$$\frac{\partial Power(\mathscr{X})}{\partial \mathscr{X}_{uv}} = \begin{bmatrix} \frac{\partial Power(\mathscr{X})}{\partial Tx_{uv}} & \frac{\partial Power(\mathscr{X})}{\partial P_{uv}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \tag{4.17}$$

Hence

$$[H_p] = \frac{\partial^2 Power(\mathscr{X})}{\partial \mathscr{X}_{uv}^2} = \begin{bmatrix} \frac{\partial^2 Power(\mathscr{X})}{\partial^2 Tx_{uv}} & \frac{\partial^2 Power(\mathscr{X})}{\partial P_{uv} \partial Tx_{uv}} \\ \frac{\partial^2 Power(\mathscr{X})}{\partial Tx_{uv} \partial P_{uv}} & \frac{\partial^2 Power(\mathscr{X})}{\partial^2 P_{uv}} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{4.18}$$

Evidently $[H_p]$ is positive semi-definite. Hence it can be concluded that $Power(\mathscr{X})$ is convex function. □

Lemma 4.4 *For a feasible transmission scenario constraints in Eq. (4.11) is differentiable under $\mathscr{X}_{uv}$ and convex.*

Proof:

$$\frac{\partial P_{uv}^{(t)}}{\partial P_{uv}} = \frac{1}{\frac{\partial P_{uv}}{\partial P_{ij}^{(t)}}} = 1$$

$$\frac{\partial X_{ijh}^{(t)}}{\partial Tx_{uv}} = \frac{1}{r_h \sigma}$$

Under feasible transmission scenario system of constraints Eq. (4.11) presents two types of non-redundant constraints.

$$\Phi[X_{ijh}^{(t)} - 1] - G_{ij} P_{ij}^{(t)} + \gamma(r_h) \sum_{fs} G_{fj} P_{fs}^{(t)} + \gamma(r_h)\eta \leq 0 \tag{4.19}$$

$$\Phi[X_{kqh}^{(t)} - 1] - G_{kr} P_{kq}^{(t)} + \gamma(r_h) G_{iq} P_{ij} + \gamma(r_h) \sum_{fs \neq ij} G_{fq} P_{fs}^{(t)} + \gamma(r_h)\eta \leq 0 \tag{4.20}$$

Let L.H.S of the Eq. (4.19), (4.20) be denoted as $SINR(\mathscr{X})$ and $SINR'(\mathscr{X})$ respectively. Hence

$$\frac{\partial SINR(\mathscr{X})}{\partial P_{uv}} = \frac{\partial SINR(\mathscr{X})}{\partial P_{uv}^{(t)}} \frac{\partial P_{uv}^{(t)}}{\partial P_{uv}} = -G_{uv}$$

Similarly

$$\frac{\partial SINR'(\mathscr{X})}{\partial P_{uv}} = \gamma(r_h)G_{uq}, \qquad \frac{\partial SINR(\mathscr{X})}{\partial Tx_{uv}} = \frac{\Phi}{r_h\sigma}, \qquad \frac{\partial SINR'(\mathscr{X})}{\partial Tx_{uv}} = 0$$

$$\frac{\partial SINR(\mathscr{X})}{\partial \mathscr{X}_{uv}} = \begin{bmatrix} \frac{\partial SINR(\mathscr{X})}{\partial Tx_{uv}} & \frac{\partial SINR(\mathscr{X})}{\partial P_{uv}} \end{bmatrix} = \begin{bmatrix} \frac{\Phi}{r_h\sigma} & -G_{uv} \end{bmatrix} \tag{4.21}$$

$$\frac{\partial SINR'(\mathscr{X})}{\partial \mathscr{X}_{uv}} = \begin{bmatrix} \frac{\partial SINR'(\mathscr{X})}{\partial Tx_{uv}} & \frac{\partial SINR'(\mathscr{X})}{\partial P_{uv}} \end{bmatrix} = \begin{bmatrix} 0 & \gamma(r_h)G_{uq} \end{bmatrix} \tag{4.22}$$

□

By the use of Lemma 4.2,4.3 and 4.4 it can be said that the Problem 4.1 is itself a multi-objective convex optimization problem. Hence according to [34] $\mathscr{X}^*$ is a Pareto optimum solution of Problem 4.1 iff the following Eq. (4.23)-(4.25) has non-negative solution for $\forall i : \lambda_i$.

$$\begin{aligned} \lambda_1 \frac{\partial Schedule(\mathscr{X}^*)}{\partial \mathscr{X}_{uv}} + \lambda_2 \frac{\partial Power(\mathscr{X}^*)}{\partial \mathscr{X}_{uv}} + \lambda_3 \frac{\partial SINR(\mathscr{X}^*)}{\partial \mathscr{X}_{uv}} \\ + \sum_q \lambda_4 q \frac{\partial SINR'(\mathscr{X}^*)}{\partial \mathscr{X}_{uv}} = \begin{bmatrix} 0 & 0 \end{bmatrix} \end{aligned} \tag{4.23}$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \sum_q \lambda_4 q = 1 \tag{4.24}$$

$$\lambda_i \geq 0 \tag{4.25}$$

Hence Eq. (4.23), (4.24) and (4.25) reduces to

$$\lambda_1 \mathfrak{P}_{uv} \left(\frac{\Gamma(\alpha)}{Tx_{uv}} + \frac{1-\Gamma(\alpha)}{Tx_{uv}^{\alpha}} \right) - \lambda_3 \frac{\Phi}{r_h\sigma} = 0 \tag{4.26}$$

$$\lambda_2 - \lambda_3 G_{uv} + \sum_q \lambda_{4q} \gamma(r_h) G_{uq} = 0 \tag{4.27}$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \sum_q \lambda_{4q} = 1 \tag{4.28}$$

For the sake of simplicity consider $\forall q : \lambda_{4q} = \lambda_4'$ and $\sum_q \lambda_{4q} = \lambda_4$ which further simplifies Eq. (4.26)-(4.28) into the following

Problem 4.2

$$\lambda_1 \mathfrak{P}_{uv} \left(\frac{\Gamma(\alpha)}{Tx_{uv}} + \frac{1-\Gamma(\alpha)}{Tx_{uv}^{\alpha}} \right) = \lambda_3 \frac{\Phi}{r_h\sigma} \tag{4.29}$$

$$\lambda_2 + \gamma(r_h)\lambda_4' \sum_q G_{uq} = \lambda_3 G_{uv} \tag{4.30}$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1 \tag{4.31}$$

In Problem 4.2 Tx_{uv}, G_{uv} and the rest of the constants are positive. Hence it can be shown that there exists a value assignment for $\lambda_1, \lambda_2, \lambda_3, \lambda_4'$ such that they are non-negative.

According to discussion above there exists a Pareto optimal solution for Problem 4.1. However as mentioned earlier Problem 4.1 is $\mathcal{NP}$-hard and by virtue of reduction so is Problem 4.2. Hence in the next section a heuristic based distributed algorithm is proposed for Fair-JPRS problem.

Chapter 5

Design of Distributed Heuristic Protocol

Fair-JPRS problem (Problem 4.1) is a $\mathcal{NP}$-hard problem, a heuristic protocol is proposed in this chapter. MCCA protocol is augmented so that it can provide fairness. Based on the Problem 4.2 each mesh STA can heuristically calculate the required transmission slots.

5.1 Proposed Protocol

Each mesh STA does the transmit power and data rate control by sensing the channel condition. Based on the interference level of the channel each mesh STA is granted some slots(TXOP). However it is very hard to know for a particular mesh STA that what would be the interference level for the receiver at the time of transmission. At the start of the protocol each station uses a test signal with P_{max} to serve the purpose of channel condition assessment. A STA v stores SINR of this received frame from a Mesh Neighbor u in a variable $\mathcal{S}_{uv}$. For this protocol to work effectively with MCCA advertisement frames $\mathcal{S}_{uv}$ measured in dB is piggybacked. Before sending a frame u can always use this piggybacked information to get an estimate of the interference in v by using the following equation which is derived from Eq. (4.12).

$$\sum_{q} G_{qv} = \frac{1}{P_{max}} \left(\frac{G_{uv} P_{max}}{\mathcal{S}_{uv}} - \eta \right) \tag{5.1}$$

Considering $\lambda_2 = \lambda_4'$, from Eq. (4.30) the following can be derived

$$\lambda_3 = \lambda_2 \frac{1 + \gamma(r_h) \sum\limits_{q} G_{uq}}{G_{uv}} \tag{5.2}$$

Considering $\eta \mathcal{S}_{uv}$ is negligible we get

$$\frac{1}{\lambda_3} \approx \frac{1}{\lambda_2} + \frac{\gamma(r_h) G_{uv}}{\lambda_2 \mathcal{S}_{uv}} \tag{5.3}$$

From Eq. (4.29) Eq. (5.4) can be obtained

$$Tx_{uv} = \begin{cases} \frac{\lambda_1 \mathfrak{P}_{uv} r_h \sigma}{\lambda_3 \Phi} & \alpha = 1 \\ (\frac{\lambda_1 \mathfrak{P}_{uv} r_h \sigma}{\lambda_3 \Phi})^{(1/\alpha)} & Otherwise \end{cases} \tag{5.4}$$

Eq. (5.1),Eq. (5.2) and Eq.(5.4) is used for solving Problem 4.2.

Each mesh STA v sends a periodic beacon frame using P_{max} at the start of DTIM interval. SINR for that frame is captured in $\mathcal{S}_{uv}$. Each mesh STA broadcasts its $\mathcal{S}_{uv}$ with MCCAOP advertisement req message. 2 Byte packet overhead for sending of SINR in DB is required. For the rate scheduling a STA requires to know r_h. For this purpose r_h is assigned with the one with highest achievable data rate. $\gamma(r_h) \leq \mathcal{S}_{uv} < \gamma(r_{h+1})$.

$$\mathcal{S}_{uv} = \frac{G_{uv} P_{max}}{\mathcal{I}} \tag{5.5}$$

$$\gamma(r_h) \leq \frac{G_{uv} P_{uv}^{(h)}}{\mathcal{I}} \tag{5.6}$$

Here $\mathcal{I}$ represents the cumulative interference and ambient noise. Using Eq. (5.5,5.6)

$$P_{uv}^{(h)} \geq \frac{\gamma(r_h)\ P_{max}}{\mathcal{S}_{uv}} \tag{5.7}$$

Though for theoretical modeling purpose this work assumes each STA can have $P_{ij}^{(t)} \in [0, P_{max}]$, in practice commodity routers come with fixed discrete power levels. In that particular case $P_{uv}^{(h)}$ is approximated to the next higher power level available in routers power level set.

Based on the solution of Problem 4.2 MCCAOP owner calculates MCCAOP parameters in the following way.

1. MCCAOP offset is set as the first free slot in the MCCAOP owner's reservation set.
2. MCCAOP Periodicity is used for ensuring short term fairness of the transmissions. To provide each mesh STA equal chance to transmit this value is set as the number of the MCCA enabled mesh neighbor(Δ).
3. If the winner does not have any prior schedule information, it assigns MCCAOP duration with a predefined fixed value Tx_{max}. If the winner already has a schedule, it estimates $\overline{\lambda}$. Based on the estimated $\overline{\lambda}$ solves Problem 4.2 by assuming Eq. (5.1). MCCAOP duration is set based on calculated $\lceil \frac{Tx_{uv}}{\Delta} \rceil$.

As per standard MCCAOP parameter along with MAF of the MCCA owner is suggested to the MCCAOP responder. A MCCAOP responder accepts the schedule if it satisfies the following conditions.

1. MAF is less than MAF limit
2. All transmission opportunity slot(TXOP)s does not overlap with the interference period of the responder.
3. All TXOP must not conflict with the neighborhood MCCAOP periods of MCCAOP owner.

If the last two cases are not satisfied, the responder re-calculates TX_{uv} from the received MCCAOP parameters and re-computes MCCAOP parameters for alternate solution. This alternate MCCAOP parameters are sent back to the MCCAOP owner via MCCAOP Setup reply frame with proper reject code. After receiving successful reservation reply from all MCCAOP responder, MCCAOP owner broadcasts MCCAOP advertisement frame to let the Mesh neighbor know about the successful reservation. Once the MCCASCANDURATION is over each mesh STA starts transmission based on the accepted schedule,data rate and transmit power.

Chapter 6

Simulation Results

The proposed scheme is validated and tested in NS-3.19 [35] network simulator. IEEE 802.11 has been considered as physical layer standard. The physical layer is capable of transmitting packets using MCS1,MCS2, MCS3 with data rate 12Mbps,18Mbps,24 Mbps and receive sensitivity of -82dbm,-80dbm, -77dbm respectively. Each node also can use 9 different power levels equally spaced between 2dbm to 17dbm. The nodes are placed in a grid structure with variable average distance between them. Based on the rest of simulation parameters listed in Table 6 the DPRL algorithm and proposed Fair-JPRS algorithm is evaluated. It is assumed that all links have similar priority($\mathfrak{P}_{uv}$).

Payload of Data Packet	512 B
Application Data Rate	10Mb/s
MCS1	Data Rate=12Mbps Receive Sensitivity=-82dbm
MCS2	Data Rate=18Mbps Receive Sensitivity=-80dbm
MCS3	Data Rate=24Mbps Receive Sensitivity=-77dbm
Min Power Level	2dbm
Max Power Level	17dbm
Power Levels	9
Slot Time σ	.80ms
DTIM	1s
Slots/DTIM	1250
Scan Duration	35ms
Slot granted to flow (ν)	50

Table 6.1 Simulation Parameters

As per discussion in the previous chapter based on Eq. (5.3), Eq. (5.4) and Eq. (5.5) the duration (Tx_{uv}) of a flow $u \to v$ inversely proportional to received SINR ($\mathcal{S}_{uv}$). The intuitive idea behind this low SINR links uses lower data rate. So for ensuring fairness they must be granted with larger number of slots. To keep the simulation simplified and comparable with existing DPRL algorithm the following method is used for calculating the Tx_{uv} by considering

$\alpha = 1$ in Eq. (5.4)

$$Tx_{uv} = \frac{\nu\psi}{\mathcal{S}_{threshold}/\mathcal{S}_{uv}} \tag{6.1}$$

Where ν denotes the number of slots granted to each flow of DPRL algorithm. $\mathcal{S}_{threshold}$ represents receive sensitivity of the MCS which is allocated to the flow. One can observe that, the parameter ψ depends on $\frac{\lambda_3}{\lambda_1}$ when the the other parameters r_h,σ,Φ remains unchanged. Thus $\psi < 1$ signifies priority over Fairness objective (Eq. (4.4)) mentioned in Problem 4.1. However $\psi > 1$ signifies priority over reduction in power consumption (4.5). Effects of change of ψ is investigated in following section.

6.1 Effect of ψ over Network Performance

Simulation run for visualizing effects of ψ is done by keeping average node distance $100m$ and 30 flows in the system. The Fig. 6.1 represents the change in power requirement when tunning is

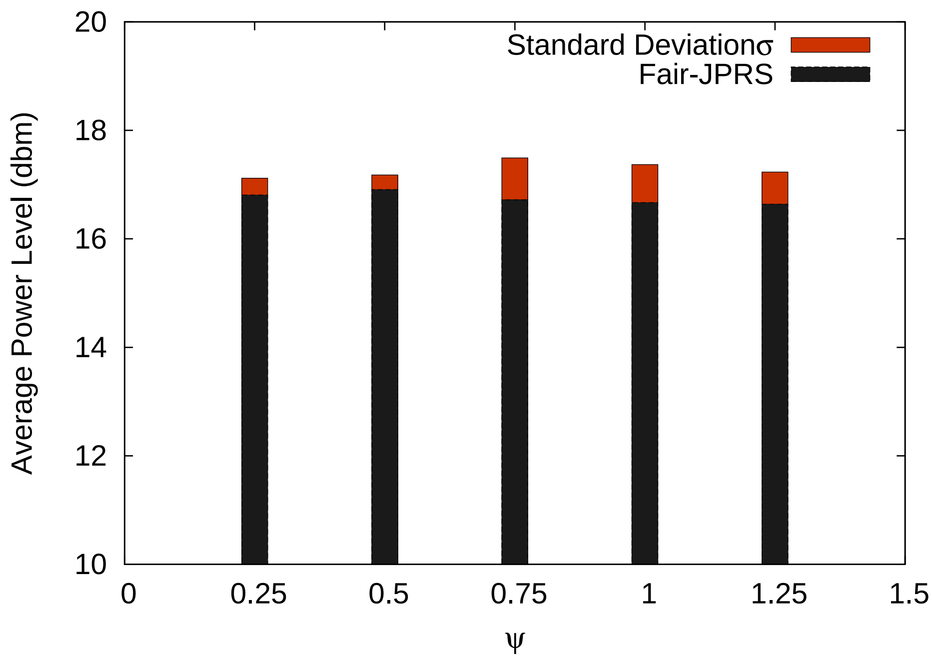

Fig. 6.1 Effect on Average Power Requirement for Different ψ

done over ψ. As mentioned earlier with increase in ψ the required average power level decreases. Fig. (**??** represents the proportional fairness index which is $\frac{\sum(\mathfrak{P}_{ij}Tx_{ij})}{\sum Tx_{ij}}$. As evident from the graph Fig. (6.3) captures the trade-off becomes much more visible. For a lower ψ one can see the huge improvement in Fairness. Here fairness of the system is calculated as given by Jain's index [36].

$$F(x) = \frac{(\sum x_i)^2}{n_f(\sum x_i^2)} \tag{6.2}$$

Where x_i denotes the throughput for flow i and n_f is the number of such flows.

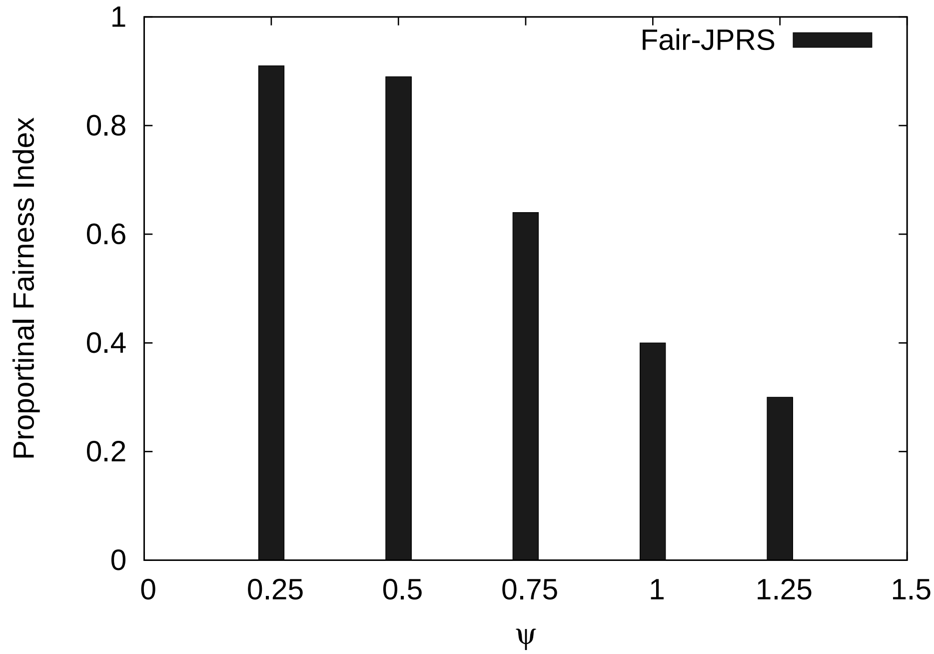

Fig. 6.2 Effect on Proportional Fairness Index for Different ψ

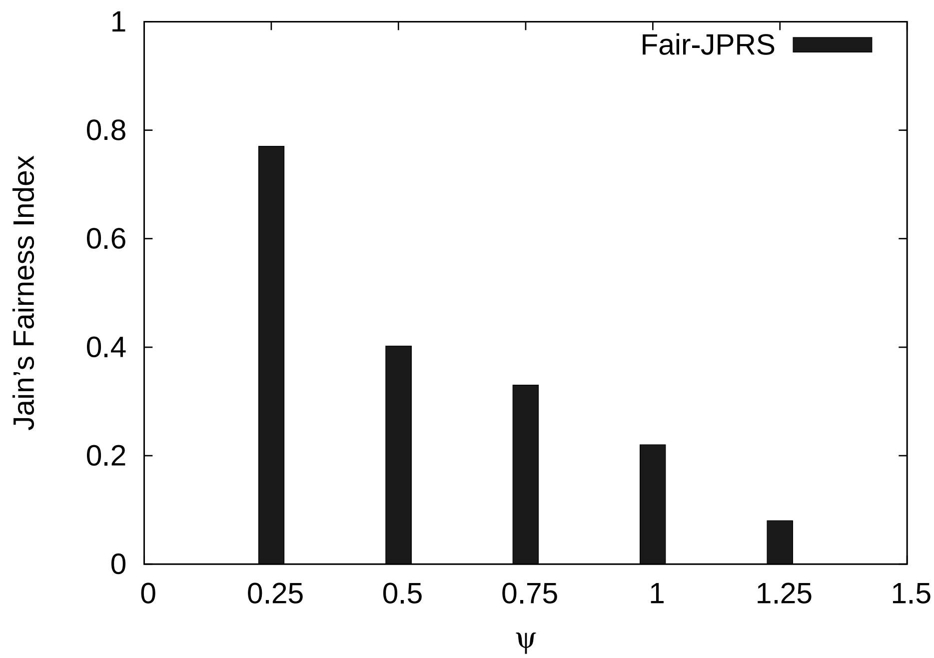

Fig. 6.3 Effect on Jain's Fairness Index for Different ψ

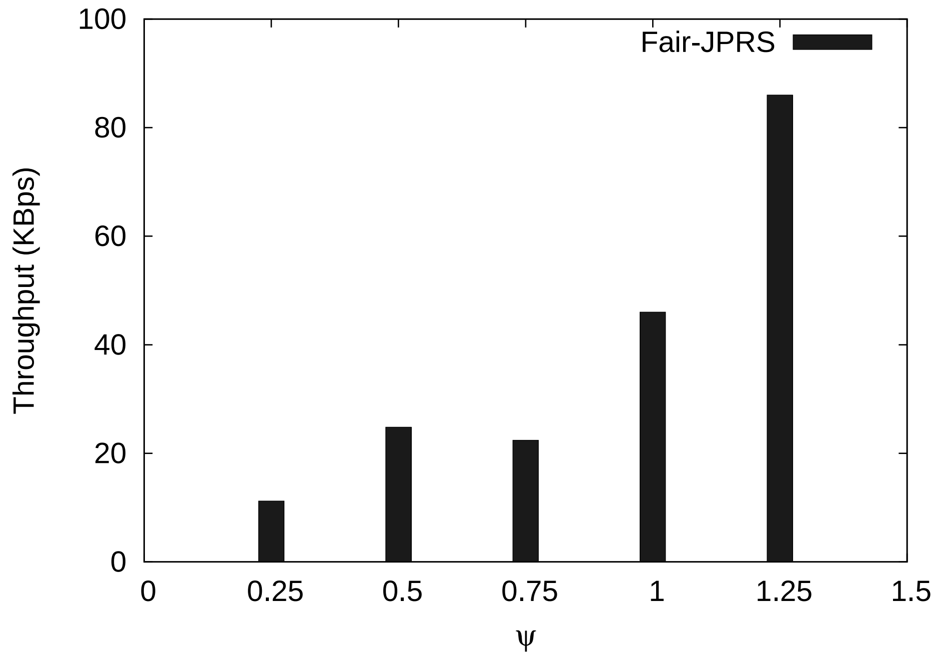

Fig. 6.4 Effect on Throughput for Different ψ

Effects of ψ over throughput shows an interesting property in Fig. 6.4. Throughput increases when $\psi = .5$. At $\psi = .75$ throughput decreases due to the fact that used power level is lowered by the protocol. So the effect of interference leads to packet drop. However if ψ is increased again the throughput rises. As most of the links do not get schedule slot, the throughput increases as the effect of interference decreases. One important observation in this case is for $\psi = .5$ both fairness and throughput shows desired behavior. Based on the fact for rest of the experiments $\psi = .5$ has been considered.

6.2 Effect of Number of nodes

To simulate the behavior of the algorithm for variable flow demands, average node distances are kept fixed as $100m$. Then the experiments are re-executed for different number of nodes in the system.

Fig. 6.5 shows the change in average power level and compares it with that of DPRL for different number of nodes. For increment in number of nodes proposed protocol shows increase in average power level due to increase in interference. Whereas DPRL shows negligible change in average power level.

The effect of increasing flows can degrade the fairness index. Fig. 6.7 shows that initially Fair-JPRS losses fairness with increase in number of flows. As with the increase in number of flows the total area of transmission increases. So simultaneous transmissions become possible. Which increases the fairness index of the Fair-JPRS. However a contrasting behavior can be seen in case of DPRL. As the interference increases SINR level drops. So the flows with higher power level gets selected very often. So the fairness of the system decreases gradually in case DPRL. However in all the cases it is observed that the fairness index for Fair-JPRS is much greater than DPRL.

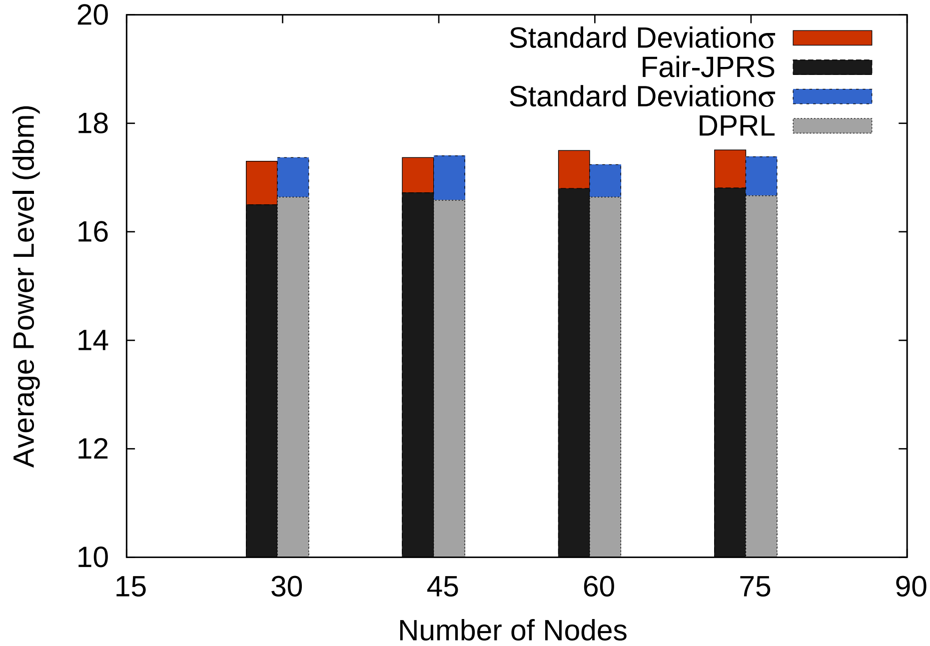

Fig. 6.5 Comparison of Fair-JPRS and DPRL Power Requirement for Variable Nodes

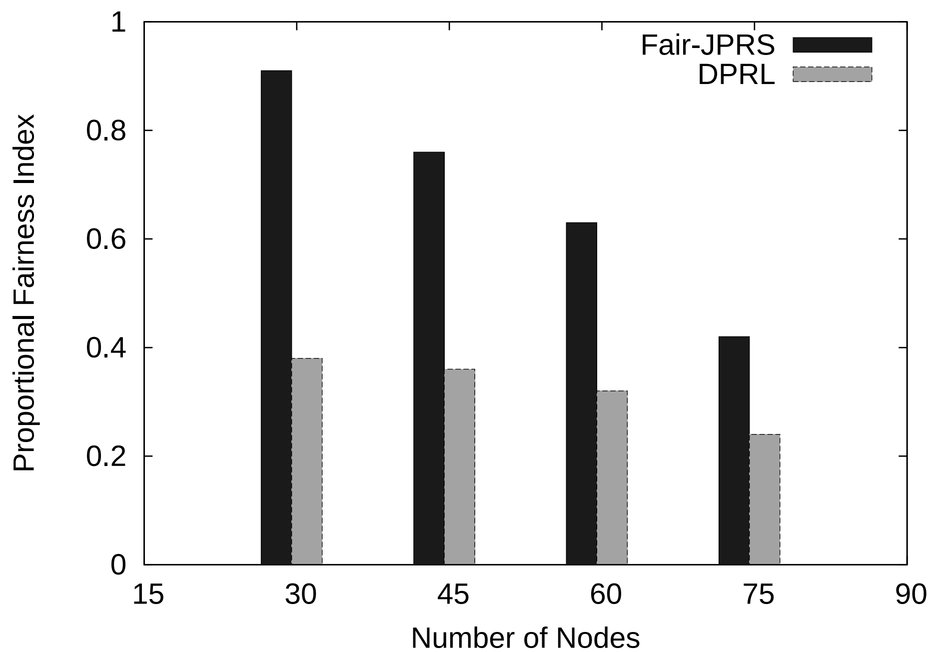

Fig. 6.6 Comparison of Proportional Fair-JPRS and DPRL Jain's Fairness Index for Variable Nodes

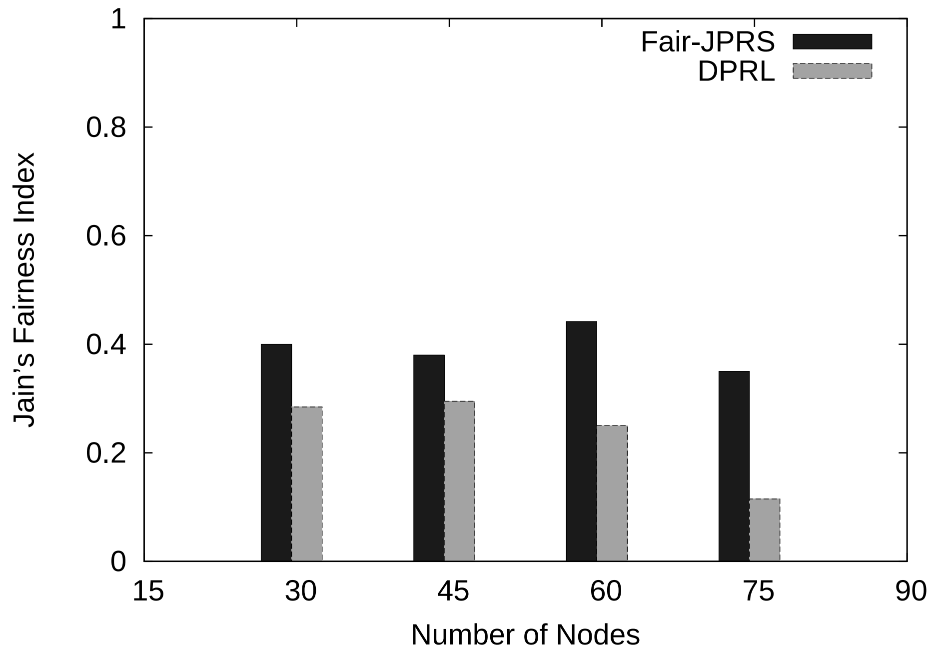

Fig. 6.7 Comparison of Fair-JPRS and DPRL Jain's Fairness Index for Variable Nodes

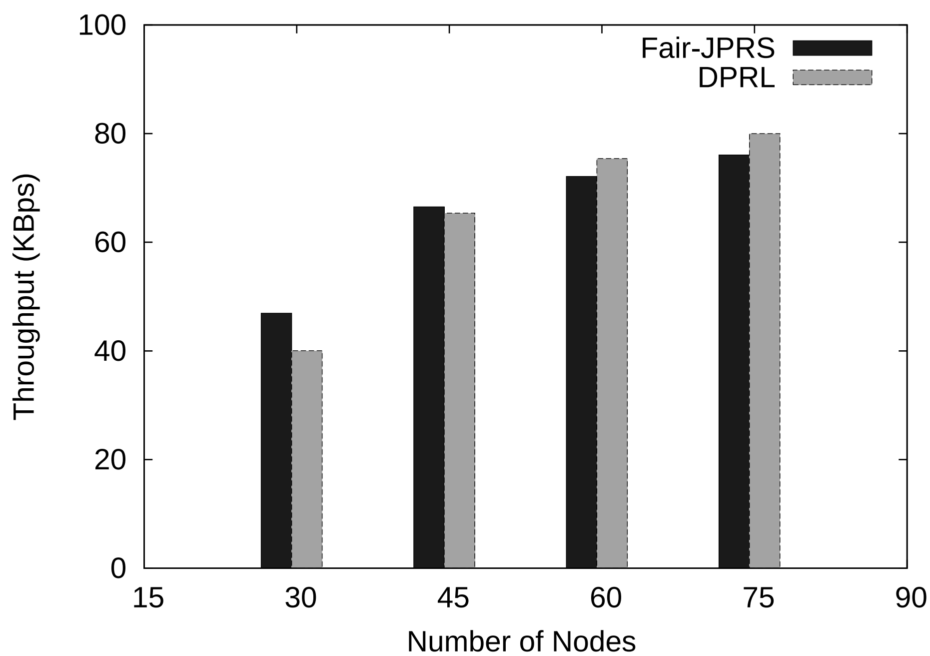

Fig. 6.8 Comparison of Fair-JPRS and DPRL Throughput for Variable Nodes

The effect of throughput over scalability is captured in Fig. 6.8. In case of DPRL the overall network throughput increases as number of nodes are added to the system. The proposed Fair-JPRS algorithm also shows approximately similar nature.

From the discussion and results it can be conclude that Fair-JPRS can performs better in terms fairness and almost similarly in terms of overall network throughput.

6.3 Effect of Average Distance Between Nodes

To perform these experiments 30 nodes have been distributed with a variable average distance between nodes.

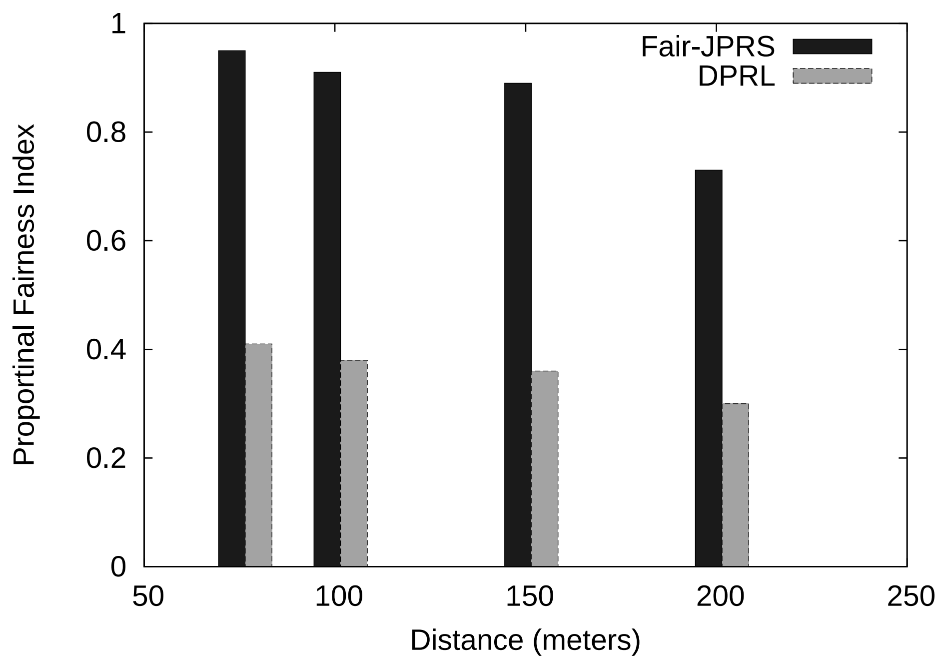

Fig. 6.9 Comparison of Fair-JPRS and DPRL Proportional Fairness Index for Variable Average Node Distance

The graphs in Fig. 6.10 shows that the increase in fairness for both Fair-JPRS and DPRL if average node distance increases. As the nodes get more distance apart the effect of interference also reduces significantly. However in this case also Fair-JPRS is more fair than DPRL.

In case of throughput Fair-JPRS and DPRL both show similar nature. When the nodes are very low distance apart the only packets gets lost only if they collied with each other. Thus increse in throughput can be observed in case of contention free channel access. However decrease in throughput can be seen when the distance between average node is $100m$ in Fig. 6.11. As the node distance increases effect of cumulative interference increases. Thus a decrease in throughput can be observed. However if the node place ments are more sparse then effect of interference also dies due to channel fading. When nodes are more than $150m$ apart, the effect of interference is so negligible. So more simultaneous transmissions are possible. This effect increases fairness and throughput for both DPRL and Fair-JPRS.

In this section the simulation results of Fair-JPRS protocol are compared with existing DPRL. The comparison shows that Fair-JPRS marginally affects overall network throughput. Average power level required for transmission remains almost unaffected. Also scalability of the

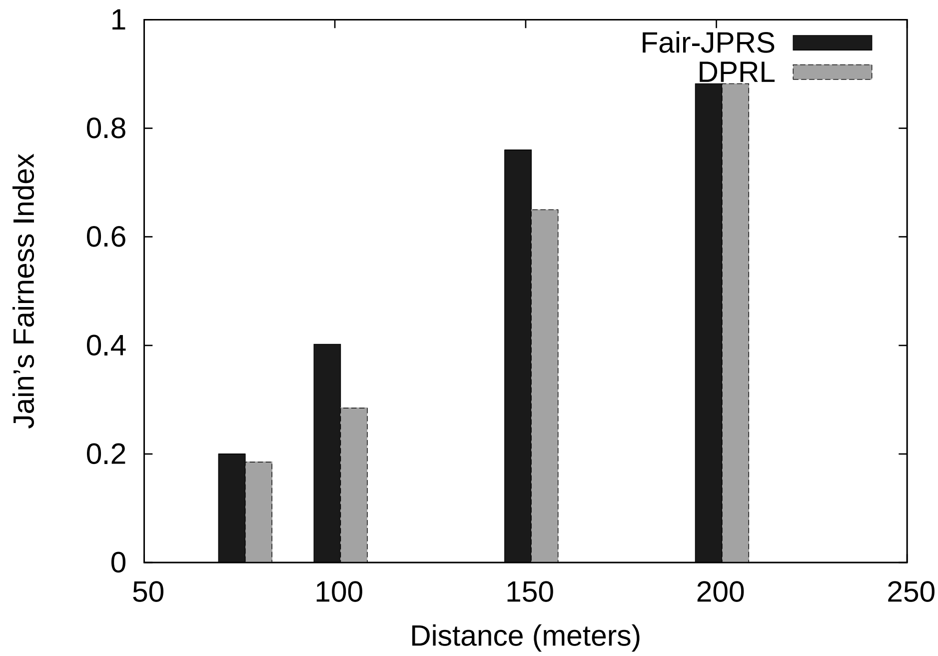

Fig. 6.10 Comparison of Fair-JPRS and DPRL Jain's Fairness Index for Variable Average Node Distance

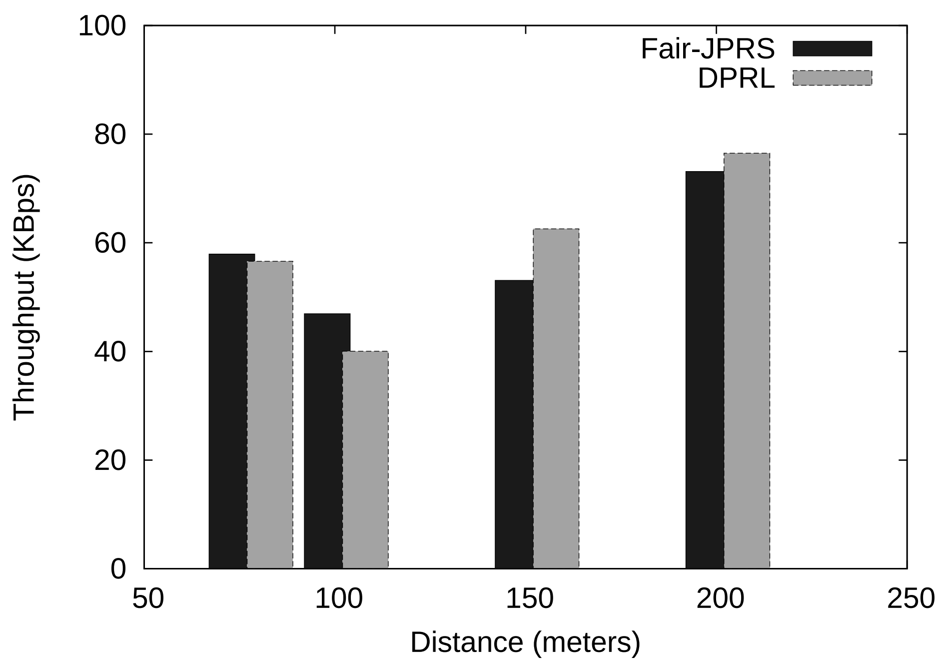

Fig. 6.11 Comparison of Fair-JPRS and DPRL Throughput for Variable Average Node Distance

protocol is ensured via simulation results. However the fairness of the system in terms of Jain's fairness index improves significantly. The results validates the acceptability of proposed scheme.

Chapter 7

Future Work and Conclusion

Network performance in terms of throughput optimization is a challenge for wireless mesh network. In this thesis a fair joint transmit power and rate scheduling problem has been studied in context of IEEE 802.11s wireless mesh network. For ensuring effective power allocation and throughput fairness a vector optimization problem is formulated. It has been proved in this context the problem is an convex optimization problem. With the help of this result existence of Pareto optimal solution is proved. However the problem being a $\mathcal{NP}$-hard problem, a heuristic is proposed in this work. The heuristic proposal is augmented with the IEEE 802.11s standard MCCA. Finally the performance of the scheme is evaluated with the help of simulation results. The simulation reveals that the augmented scheme performs well without significant loss of throughput. However the fairness improvement of proposed protocol shows significant improvement over existing work.

For the sake of simplicity the proposed scheme is developed for single channel single interface wireless mesh network with omni-directional antenna. However the scheme can be extended for multi channel and multi interface network also. Multi class traffic can be supported in the extended version of this scheme. Furthermore, a future study can be done to extend the work for directional antenna support. Another interesting point of study could be further improvement in throughput without affecting fairness criteria.

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